

## BINGOL, Turkey

WMO: 172030

Lat: 38.885N

Lon: 40.501E

Elev: 1177

StdP: 87.96

Time Zone: 3.00 (E03)

Period: 98-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -13.3      | -10.8 | -17.2                         | 1.0 | -10.4 | -15.5 | 1.1 | -8.8 | 4.2                   | -0.2 | 3.4 | -0.4 | 0.1                   | 310  | 0.417 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 14.3                   | 36.4            | 19.9 | 35.1 | 19.3 | 33.8 | 18.8 | 21.8                | 34.5 | 20.8 | 33.1 | 19.9 | 31.9 | 1.2                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 17.2                            | 14.2 | 30.9 | 16.0 | 13.1 | 30.2 | 15.0 | 12.3 | 29.1 | 69.7          | 34.8 | 65.9 | 33.1 | 62.5 | 31.9 | 25.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 3.9           | 3.2  | 2.5 | -14.9                      | 38.7 | 4.1                | 1.5 | -17.8                                              | 39.7 | -20.2      | 40.6 | -22.5      | 41.4 | -25.4      | 42.5 |    |    |
| (5)               |      |     | -15.6                      | 23.0 | 3.6                | 1.5 | -18.2                                              | 24.1 | -20.4      | 25.0 | -22.4      | 25.8 | -25.0      | 26.9 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan    | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)    | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 12.6   | -1.8  | -0.3  | 5.3   | 10.9  | 16.2  | 22.3  | 26.8  | 26.9  | 21.4  | 15.1  | 6.9   | 0.7  | (6)  |
| (7)  |                                                                                  | DBStd       | 10.66  | 4.25  | 4.31  | 4.30  | 3.91  | 3.47  | 2.86  | 2.50  | 1.97  | 2.99  | 3.63  | 4.07  | 4.67 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 1244   | 366   | 288   | 156   | 35    | 2     | 0     | 0     | 0     | 0     | 6     | 103   | 288  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 2864   | 624   | 521   | 404   | 223   | 81    | 4     | 0     | 0     | 9     | 107   | 342   | 546  | (9)  |
| (10) |                                                                                  | CDD10.0     | 2202   | 0     | 0     | 9     | 62    | 196   | 369   | 522   | 524   | 343   | 165   | 11    | 1    | (10) |
| (11) |                                                                                  | CDD18.3     | 781    | 0     | 0     | 0     | 1     | 16    | 124   | 263   | 266   | 103   | 8     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 9004   | 0     | 0     | 0     | 14    | 184   | 1316  | 3199  | 3136  | 1044  | 111   | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 4353   | 0     | 0     | 0     | 1     | 23    | 501   | 1720  | 1702  | 392   | 14    | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 0.6    | 0.3   | 0.3   | 0.7   | 0.6   | 0.8   | 0.9   | 0.7   | 0.6   | 0.4   | 0.5   | 0.3   | (14) |      |
| (15) | Precipitation                                                                    | PrecAvg     | 845    | 115   | 125   | 119   | 106   | 75    | 22    | 5     | 15    | 62    | 93    | 110   | (15) |      |
| (16) |                                                                                  | PrecMax     | 1099   | 355   | 278   | 228   | 250   | 217   | 69    | 25    | 25    | 78    | 205   | 207   | 234  | (16) |
| (17) |                                                                                  | PrecMin     | 554    | 10    | 49    | 36    | 32    | 9     | 2     | 0     | 0     | 0     | 6     | 3     | 5    | (17) |
| (18) |                                                                                  | PrecStd     | 136    | 69    | 51    | 48    | 51    | 43    | 15    | 6     | 6     | 18    | 50    | 54    | 64   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 8.5   | 12.0  | 19.6  | 25.1  | 28.9  | 34.0  | 38.5  | 37.7  | 34.2  | 28.4  | 19.8  | 13.9 | (19) |
| (20) |                                                                                  |             | MCWB   | 3.6   | 5.5   | 10.8  | 12.5  | 15.6  | 18.3  | 21.1  | 20.8  | 19.4  | 15.1  | 10.5  | 7.7  | (20) |
| (21) |                                                                                  | 2%          | DB     | 6.2   | 9.1   | 16.9  | 22.6  | 27.1  | 32.4  | 36.8  | 36.4  | 32.1  | 26.1  | 17.3  | 10.2 | (21) |
| (22) |                                                                                  |             | MCWB   | 2.6   | 4.5   | 9.1   | 11.7  | 15.2  | 18.3  | 20.0  | 19.9  | 17.9  | 14.2  | 9.7   | 5.1  | (22) |
| (23) |                                                                                  | 5%          | DB     | 4.9   | 7.1   | 14.7  | 20.2  | 25.5  | 31.1  | 35.3  | 35.3  | 30.6  | 24.2  | 15.4  | 8.0  | (23) |
| (24) |                                                                                  |             | MCWB   | 2.3   | 3.4   | 7.9   | 10.8  | 14.4  | 17.8  | 19.5  | 19.3  | 17.2  | 13.8  | 8.5   | 4.5  | (24) |
| (25) |                                                                                  | 10%         | DB     | 3.5   | 5.2   | 12.3  | 18.1  | 23.7  | 29.8  | 34.0  | 34.0  | 29.1  | 22.2  | 13.2  | 6.3  | (25) |
| (26) |                                                                                  |             | MCWB   | 1.3   | 2.4   | 7.0   | 10.2  | 13.9  | 17.3  | 19.0  | 18.7  | 16.6  | 13.2  | 7.3   | 3.7  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 5.1   | 6.6   | 11.7  | 13.9  | 17.3  | 20.7  | 23.3  | 23.1  | 20.6  | 16.7  | 11.5  | 8.2  | (27) |
| (28) |                                                                                  |             | MCD B  | 6.4   | 10.1  | 17.5  | 21.5  | 26.4  | 31.7  | 36.3  | 35.1  | 32.0  | 26.3  | 17.9  | 11.3 | (28) |
| (29) |                                                                                  | 2%          | WB     | 3.7   | 5.0   | 10.0  | 12.5  | 16.1  | 19.6  | 22.1  | 21.8  | 19.5  | 15.5  | 10.1  | 6.2  | (29) |
| (30) |                                                                                  |             | MCD B  | 5.2   | 8.2   | 15.6  | 20.1  | 25.3  | 30.7  | 34.8  | 34.5  | 30.5  | 24.1  | 15.7  | 8.7  | (30) |
| (31) |                                                                                  | 5%          | WB     | 2.6   | 3.9   | 8.7   | 11.6  | 15.1  | 18.8  | 21.1  | 20.8  | 18.4  | 14.6  | 9.2   | 5.1  | (31) |
| (32) |                                                                                  |             | MCD B  | 4.3   | 6.3   | 13.5  | 18.4  | 23.8  | 29.7  | 33.6  | 33.2  | 29.0  | 22.7  | 13.9  | 7.2  | (32) |
| (33) |                                                                                  | 10%         | WB     | 1.6   | 2.8   | 7.3   | 10.6  | 14.2  | 17.8  | 20.1  | 20.0  | 17.3  | 13.6  | 8.2   | 4.1  | (33) |
| (34) |                                                                                  |             | MCD B  | 3.2   | 5.0   | 11.7  | 16.8  | 22.3  | 28.6  | 32.4  | 32.1  | 27.8  | 21.1  | 12.2  | 6.0  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | MDBR        | 6.8    | 7.5   | 9.0   | 10.1  | 12.0  | 13.6  | 14.3  | 14.9  | 14.7  | 12.4  | 10.2  | 7.5   | (35) |      |
| (36) |                                                                                  | 5% DB       | MCD BR | 8.2   | 9.5   | 12.7  | 13.9  | 14.5  | 14.6  | 15.4  | 16.1  | 16.5  | 15.5  | 13.9  | 10.1 | (36) |
| (37) |                                                                                  |             | MCWBR  | 5.4   | 6.0   | 6.8   | 6.7   | 6.6   | 7.0   | 7.2   | 7.4   | 8.1   | 7.6   | 7.6   | 6.2  | (37) |
| (38) |                                                                                  | 5% WB       | MCD BR | 6.4   | 8.1   | 10.8  | 11.9  | 13.2  | 14.1  | 14.7  | 14.9  | 15.3  | 13.9  | 12.3  | 8.6  | (38) |
| (39) |                                                                                  |             | MCWBR  | 4.4   | 5.2   | 6.0   | 6.1   | 6.5   | 7.4   | 7.7   | 7.8   | 8.4   | 7.1   | 6.9   | 5.6  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.305  | 0.340 | 0.383 | 0.438 | 0.403 | 0.356 | 0.359 | 0.362 | 0.330 | 0.371 | 0.323 | 0.303 | (40) |      |
| (41) |                                                                                  | taud        | 2.221  | 2.111 | 2.102 | 2.047 | 2.227 | 2.393 | 2.389 | 2.360 | 2.437 | 2.276 | 2.382 | 2.300 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 855    | 867   | 866   | 839   | 878   | 919   | 912   | 902   | 917   | 827   | 835   | 831   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 100    | 129   | 145   | 164   | 140   | 119   | 118   | 118   | 103   | 109   | 86    | 87    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.29   | 3.29  | 4.34  | 5.08  | 6.32  | 7.82  | 7.83  | 6.98  | 5.71  | 3.76  | 2.60  | 1.98  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.32   | 0.46  | 0.35  | 0.58  | 0.45  | 0.39  | 0.27  | 0.21  | 0.25  | 0.28  | 0.30  | 0.37  | (45) |      |

## Historical Trends

|      |                        | Heating |        | Cooling |       |       | Degree-Days |         |         |         |      |
|------|------------------------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|------|
|      |                        | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |      |
| (d)  | (e)                    | (f)     | (g)    | (h)     | (i)   | (j)   | (k)         | (l)     | (m)     |         |      |
| (46) | Station Only           | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | (46)    |      |
| (47) | Regional (6 neighbors) | N/A     | N/A    | +0.68   | -0.79 | N/A   | N/A         | N/A     | N/A     | +34     | (47) |

Nomenclature: See separate page